CreditAI — Agentic Lending Decision Support

Technical Report (Milestone 2)

GenAI Capstone · NST Sonipat · 2026

---

1. Problem Statement

Traditional lending systems rely on static scoring rules or opaque models that lack transparency. These approaches fail to:

- Clearly explain why a borrower is risky
- Align decisions with financial regulations
- Provide structured outputs for human review

This project addresses these gaps by building an **AI-powered lending decision support system** that:

- Predicts borrower default probability using a trained ML model
- Identifies and explains key risk factors
- Retrieves relevant regulatory guidelines using RAG
- Generates a structured lending recommendation report

The system is designed as a **decision-support tool**, not a replacement for human judgment.

---

2. System Overview

The system integrates multiple AI components into a single pipeline:

- **Machine Learning Model (Decision Tree)** → Predicts default risk
- **Agentic Workflow (LangGraph)** → Controls execution flow
- **RAG System (FAISS)** → Retrieves regulatory context
- **LLM (HuggingFace)** → Generates structured reports
- **Frontend (Streamlit)** → User interaction and visualization

This combination enables **accurate, explainable, and regulation-aware decision-making**.

---

3. System Architecture

The system is organized into four layers:

Layer 1 — Input (UI)

- Streamlit-based form
- Collects 17 borrower features
- Performs basic validation

Layer 2 — Agent (LangGraph)

- Stateful workflow using nodes
- Ensures sequential execution
- Maintains audit logs

Layer 3 — Intelligence

- Decision Tree model for prediction
- FAISS for regulatory retrieval
- LLM for report generation

Layer 4 — Output

- Risk score and classification
- Feature importance
- Structured downloadable report

Data Flow

1. User inputs borrower details
2. Data is validated and preprocessed
3. ML model predicts default probability
4. RAG retrieves relevant regulatory text
5. LLM generates structured report
6. Results displayed on dashboard

---

4. ML Model

Model Used

Decision Tree Classifier (from Milestone 1)

Why Decision Tree

- High accuracy (≈90.8%)

- Strong recall for default cases (≈0.77)
- Provides feature importance → explainability

Hyperparameters

Parameter	Value	Purpose
max_depth	10	Prevent overfitting
min_samples_split	20	Ensure meaningful splits
min_samples_leaf	10	Reduce variance
class_weight	balanced	Handle class imbalance

Performance

Metric	Value
Accuracy	90.81%
ROC-AUC	0.68
Precision (Default)	0.81
Recall (Default)	0.77
F1 Score	0.79

Confusion Matrix (Summary)

- True Negatives: 6006
- True Positives: 1391
- False Positives: 325
- False Negatives: 424

Interpretation:
Model prioritizes detecting defaulters, reducing financial risk.

---

5. Key Risk Drivers

Top features influencing predictions:

1. **Loan Term** → Longer tenure increases risk
2. **Debt-to-Income Ratio** → Higher ratio = financial stress
3. **Annual Income** → Lower income increases default risk
4. **Credit Lines** → Indicates credit exposure
5. **Delinquency History** → Past behavior predictor

These features are extracted using **Decision Tree feature importance**.

---

6. Agentic Workflow (LangGraph)

The system uses a **4-node workflow**:

Node 1 — Profile Analysis

- Validates input
- Handles missing values
- Logs initial state

Node 2 — ML Risk Scoring

- Loads trained model
- Generates:
  - Default probability
  - Risk label
  - Risk tier
  - Feature importance

Node 3 — Regulation Retrieval (RAG)

- Builds query from borrower profile
- Retrieves relevant documents from FAISS
- Sources include:
  - RBI guidelines
  - Basel III framework
  - Fair lending policies

Node 4 — Report Generation

- Combines:
  - Borrower data
  - ML predictions
  - Regulatory context
- Uses LLM to generate structured report

---

7. RAG System

Purpose

To ensure model outputs are **factually grounded and compliant**.

Pipeline

1. Load regulatory documents
2. Split into chunks
3. Convert to embeddings
4. Store in FAISS index
5. Retrieve top relevant chunks

Benefits

- Reduces hallucinations
- Improves explainability
- Aligns decisions with real policies

---

8. Output Structure

Each prediction generates:

Risk Metrics

- Default probability (%)
- Risk label (Low / High)
- Risk tier (Excellent → Very High)

Structured Report Sections

1. Borrower Summary
2. Risk Analysis
3. Regulatory Considerations
4. Lending Recommendation
5. Conditions (if required)
6. Disclaimer

---

9. User Interface (Streamlit)

Features

- Input form (17 features)
- KPI cards (risk score, tier)
- Feature importance visualization
- Audit logs (workflow steps)
- Regulatory sources display
- Downloadable report

---

10. Input–Output Specification

Inputs

- Demographics (age, employment)
- Financials (income, loan amount)
- Credit behavior (delinquencies, utilization)
- Loan details (term, interest rate)

Outputs

- Default probability
- Risk classification
- Top risk drivers
- Regulatory references
- Full structured report

---

11. Responsible AI

- No sensitive attributes used
- Transparent feature importance
- Regulatory grounding via RAG
- Low-temperature LLM (reduces hallucination)
- Human-in-the-loop decision system

---

12. Limitations

- Moderate ROC-AUC (0.68)
- Static dataset (no real-time updates)
- Limited fairness evaluation
- Dependency on LLM API

---

13. Deployment

- Streamlit web application
- Can be hosted on:
    - Streamlit Cloud
    - HuggingFace Spaces
    - Render

---

14. Tech Stack

- Python 3.x
- Scikit-learn
- Pandas, NumPy
- LangGraph
- FAISS
- HuggingFace API
- Streamlit
- GitHub

---

15. Conclusion

This project demonstrates a **production-style AI system** that combines:

- Machine Learning (prediction)
- Agentic workflows (control logic)
- RAG (knowledge grounding)
- LLMs (report generation)

The system delivers **accurate, explainable, and regulation-aware credit decisions**, making it suitable for real-world financial applications.